

## HW 3 - Agentic AI

### Part 0 -

SID4: 0686

PORT\_BASE: 8686

PREFIX: s0686

SEED: 686

VERIFY\_SEED: 260686

DOMAIN\_ID: 6

Domain: Rental Housing Listings

Hardware: MacBook Air

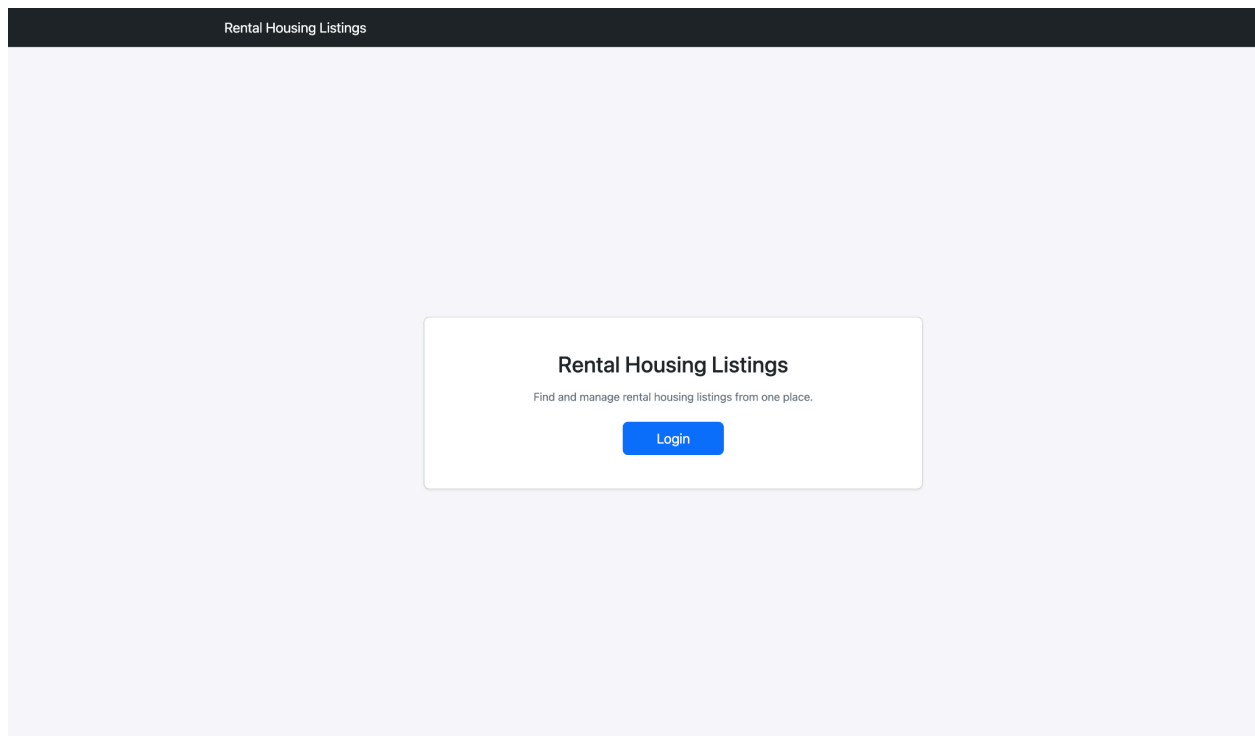
GitHub Repository: <https://github.com/farzaandalwai/data260-0686>

Local embedding model: sentence-transformers/all-M

### Part 1 screenshots -

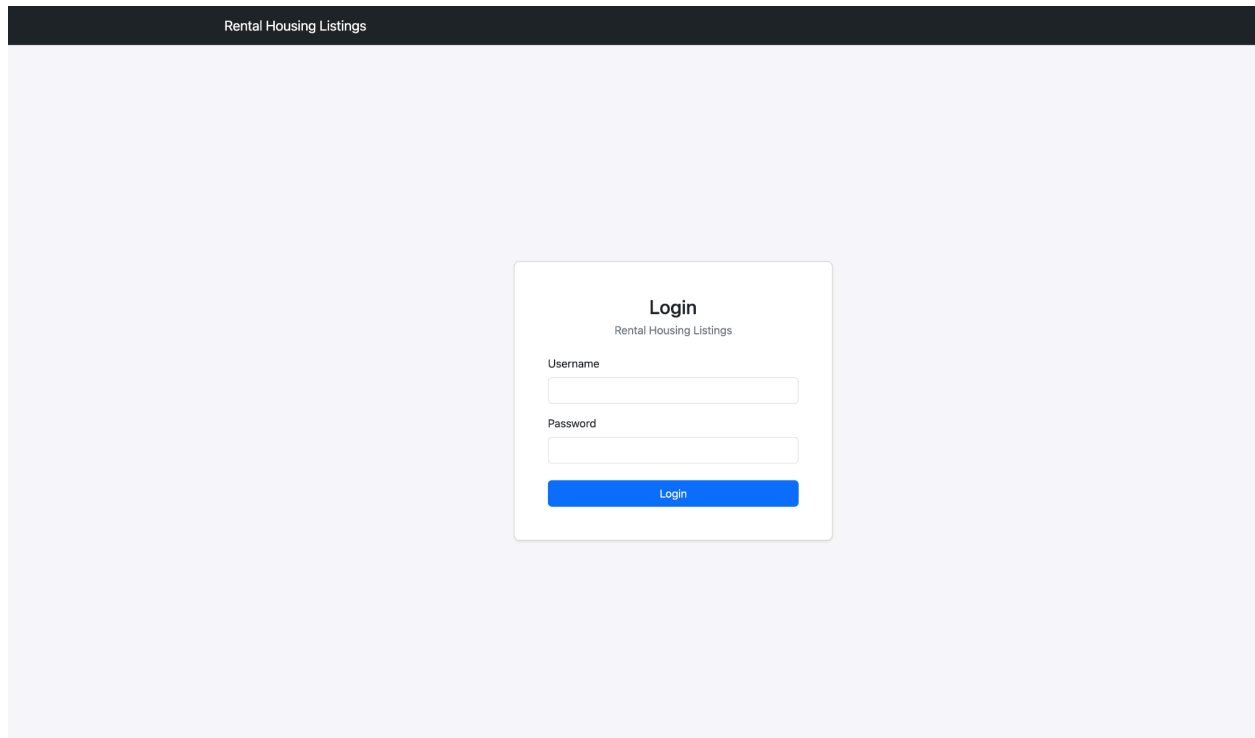
#### **Figure 1. Rental Housing Listings Home Page - Logged-Out State**

The home route (/) displays the domain-specific Rental Housing Listings interface. When no authenticated session exists, the user is presented with a Login option.



**Figure 2. Login Page**

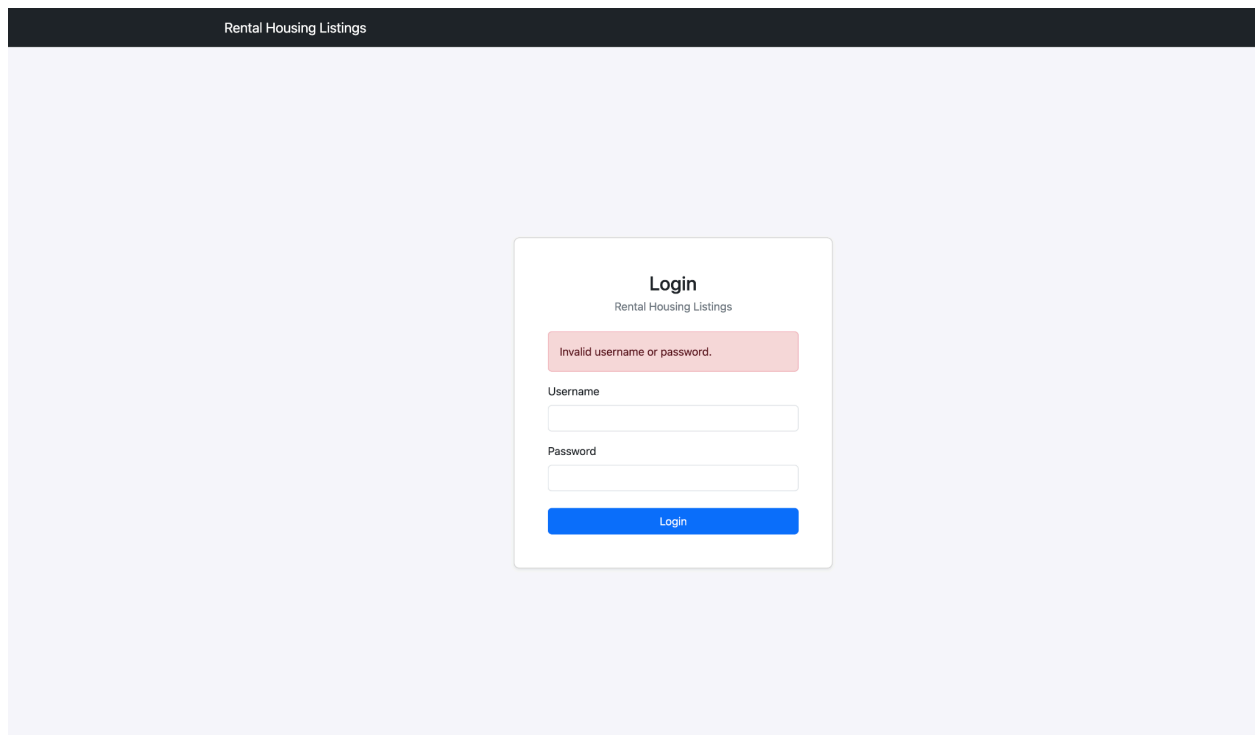
The `/login` route provides username and password fields in a Bootstrap-styled login card.



The screenshot shows a web application with a dark header bar containing the text "Rental Housing Listings". The main content area is light gray and contains a white login card. The card has a title "Login" and a subtitle "Rental Housing Listings". Below the subtitle are two input fields: "Username" and "Password". At the bottom of the card is a blue button labeled "Login".

### Figure 3. Invalid Login Validation

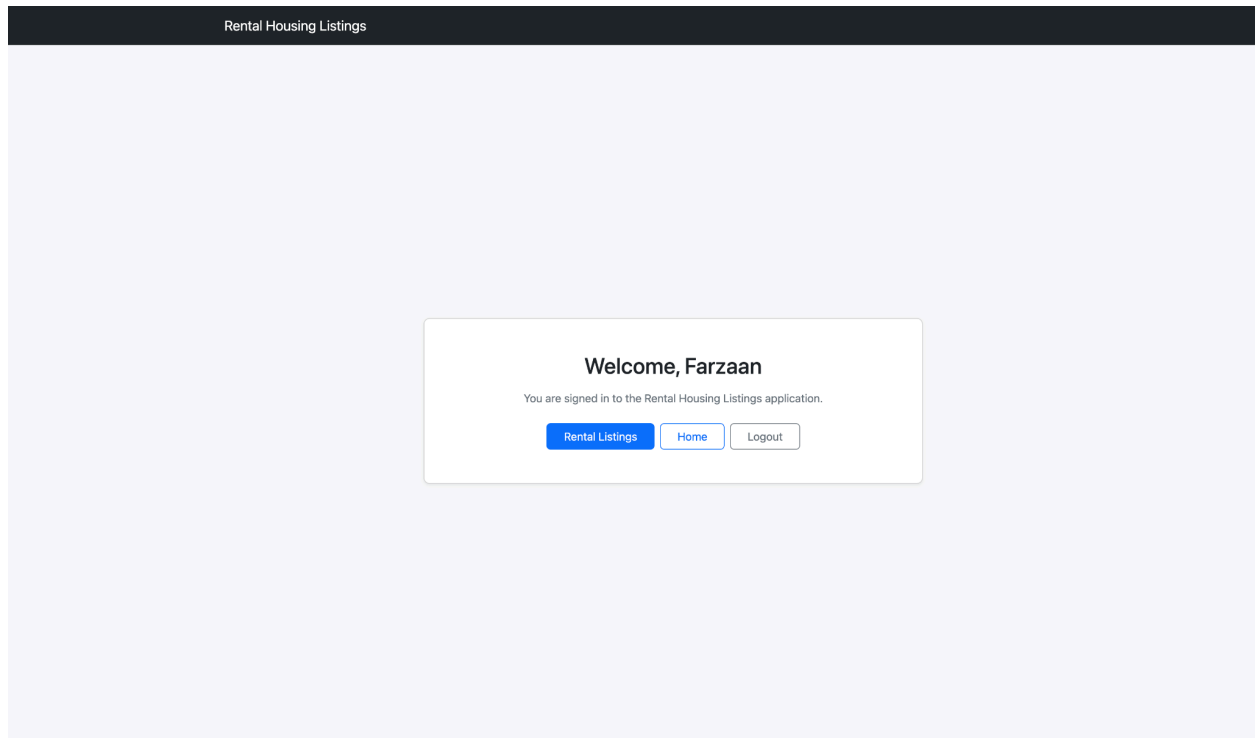
When incorrect credentials are submitted, the application remains on the login page and displays a Bootstrap danger alert stating that the username or password is invalid.



The screenshot shows the same web application as Figure 3, but with a red alert message displayed above the input fields. The alert message reads "Invalid username or password." The "Username" and "Password" input fields and the "Login" button are still visible below the alert.

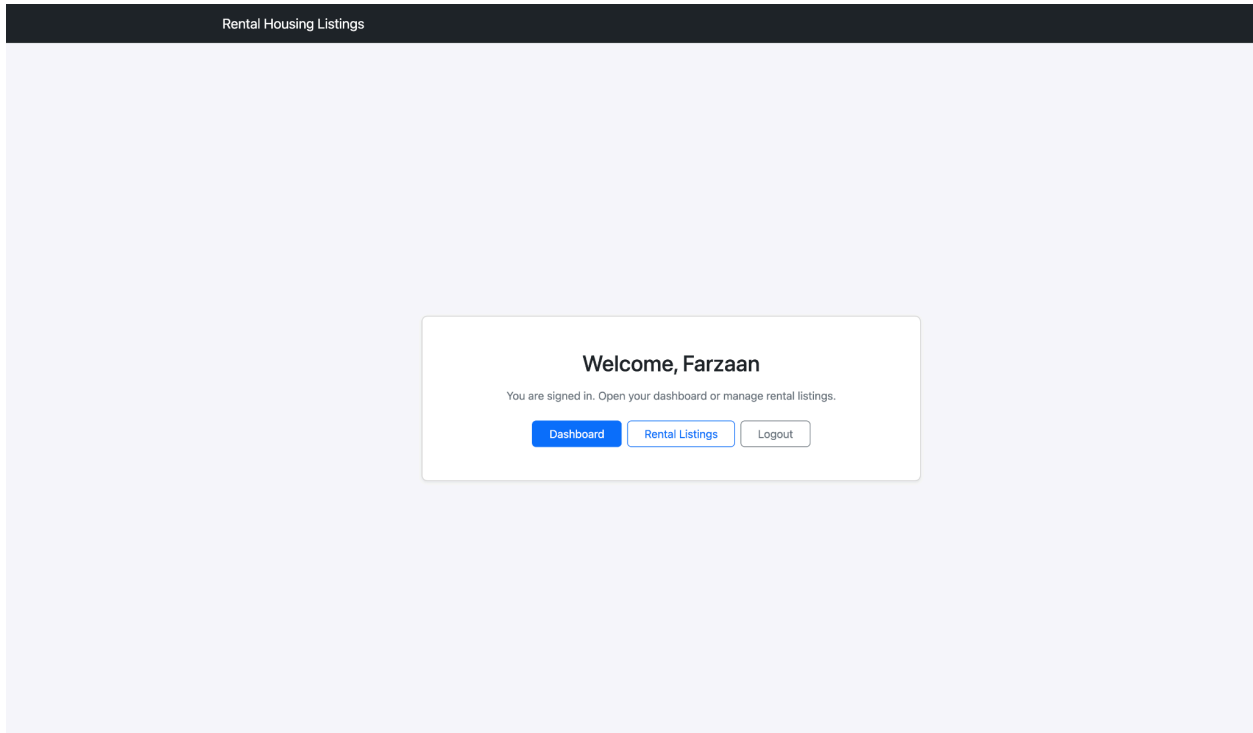
### Figure 4. Protected Dashboard After Successful Login

After successful authentication, the user is redirected to /dashboard. The protected page displays “Welcome, Farzaan” and provides access to the rental listings interface, home page, and logout action.



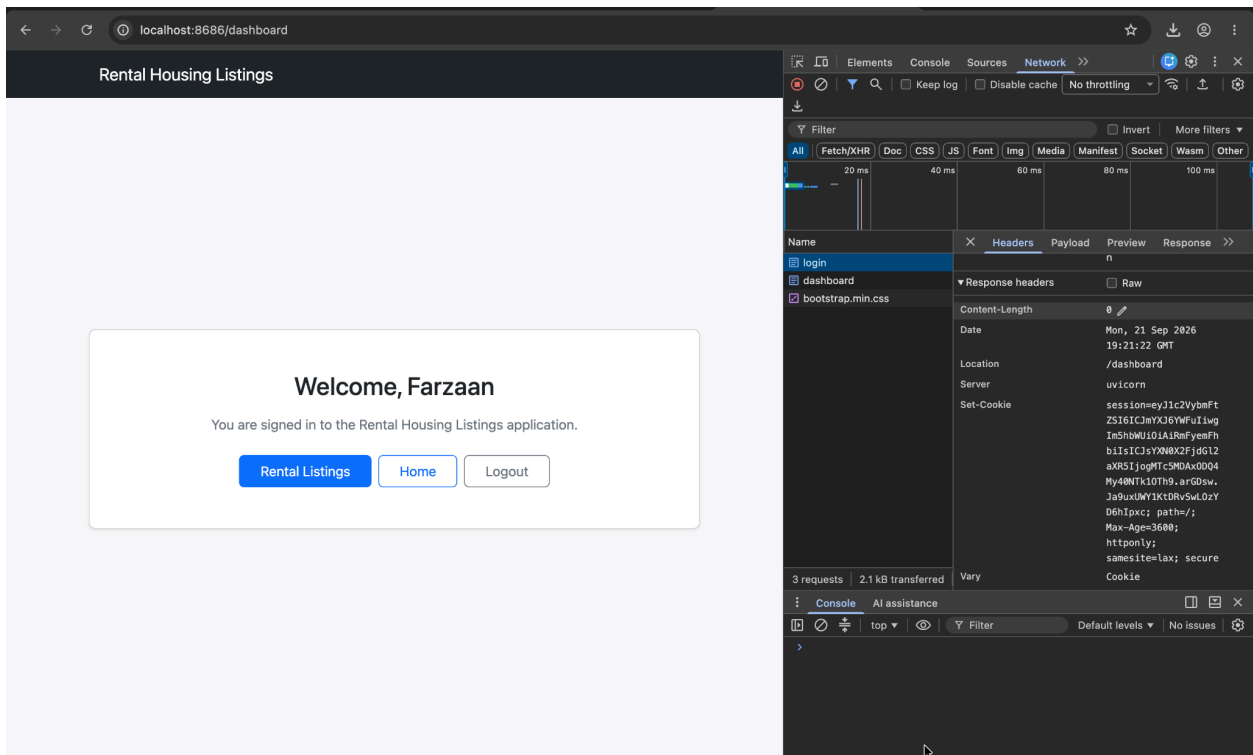
**Figure 5. Home Page Authenticated State**

When a valid session exists, the home page recognizes the authenticated user and displays Dashboard, Rental Listings, and Logout actions.



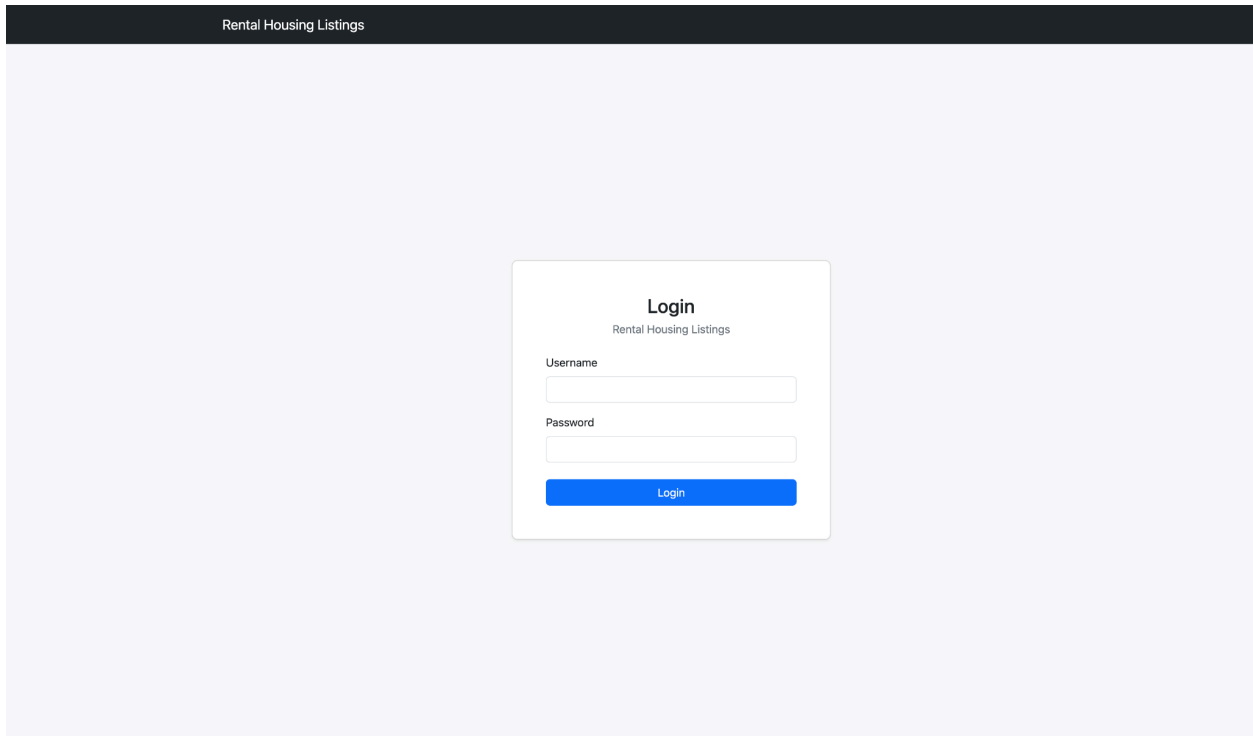
**Figure 6. Secure Session Cookie Response Header**

Chrome DevTools shows the Set-Cookie response header returned after a successful login. The session cookie includes the required HttpOnly, SameSite=lax, and Secure attributes.



**Figure 7. Dashboard Access Denied After Logout**

After the user logs out, the session is cleared. A subsequent attempt to access </dashboard> redirects the user back to the login page, demonstrating that the previous authenticated session cannot be reused.

The image shows a web application interface. At the top, there is a dark header bar with the text "Rental Housing Listings" in white. Below the header, the main content area has a light blue background. In the center of this area is a white rectangular box with rounded corners. Inside this box, the word "Login" is centered at the top in a bold font. Below it, the text "Rental Housing Listings" is centered in a smaller font. There are two input fields: the first is labeled "Username" and the second is labeled "Password". Both fields are empty. Below the password field is a blue button with the word "Login" in white text.

Verification result:

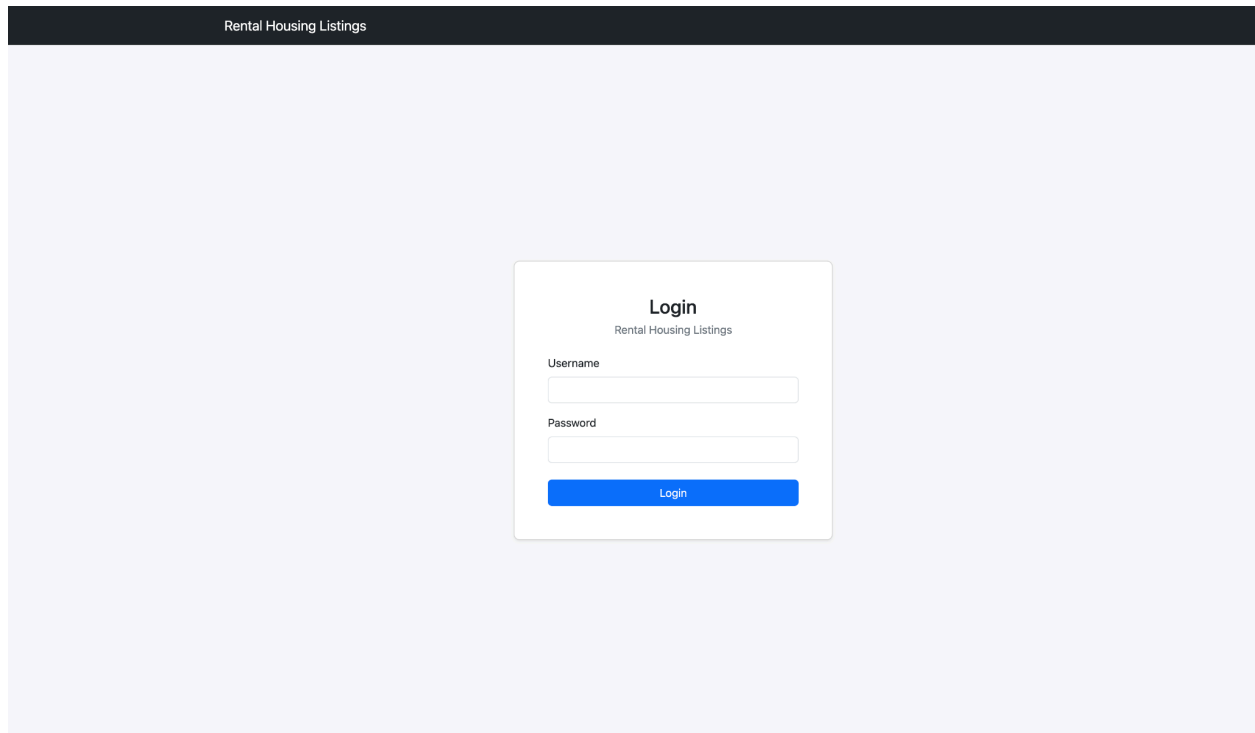
GET /dashboard after logout

HTTP 303

Location: /login

### **Figure 8. Dashboard Access Denied After Idle Session Timeout**

The application uses a 60-second idle-session timeout. After remaining idle for more than 60 seconds, a request to /dashboard redirects the user to /login, proving that the expired session can no longer access the protected route.



Idle timeout: 60 seconds

Result after 61 seconds:

GET /dashboard

HTTP 303

Location: /login

### Figure 9. Authentication Templates Directory

The FastAPI application uses separate Jinja2 templates for the home, login, and dashboard pages.

|                  |                   |      |               |
|------------------|-------------------|------|---------------|
| ▼ templates      | Today at 11:05 AM | --   | Folder        |
| 📄 dashboard.html | Today at 11:29 AM | 1 KB | HTML document |
| 📄 home.html      | Today at 11:28 AM | 2 KB | HTML document |
| 📄 login.html     | Today at 11:28 AM | 2 KB | HTML document |

The following routers/auth.py file contains the authentication routes, login validation, session checks, logout behavior, and idle-session timeout used in Part 1.

```
import time
from pathlib import Path

from fastapi import APIRouter, Form, Request
from fastapi.responses import RedirectResponse
```

```

from fastapi.templating import Jinja2Templates

router = APIRouter()
templates = Jinja2Templates(
    directory=str(Path(__file__).resolve().parent.parent / "templates")
)

USERNAME = "farzaan"
PASSWORD = "rental260"
DISPLAY_NAME = "Farzaan"
SESSION_IDLE_SECONDS = 60

def session_is_active(request):
    if not request.session.get("username"):
        return False

    last_activity = request.session.get("last_activity")
    now = time.time()
    if last_activity is None or now - float(last_activity) > SESSION_IDLE_SECONDS:
        request.session.clear()
        return False

    request.session["last_activity"] = now
    return True

@router.get("/")
def home(request: Request):
    logged_in = session_is_active(request)
    return templates.TemplateResponse(
        request,
        "home.html",
        {
            "logged_in": logged_in,
            "name": request.session.get("name", ""),
        },
    )

```

```

)

@router.get("/login")
def login_page(request: Request):
    if session_is_active(request):
        return RedirectResponse("/dashboard", status_code=303)
    return templates.TemplateResponse(
        request,
        "login.html",
        {"error": ""},
    )

```

```

@router.post("/login")
def login(
    request: Request,
    username: str = Form(),
    password: str = Form(),
):
    if username == USERNAME and password == PASSWORD:
        request.session["username"] = USERNAME
        request.session["name"] = DISPLAY_NAME
        request.session["last_activity"] = time.time()
        return RedirectResponse("/dashboard", status_code=303)

    return templates.TemplateResponse(
        request,
        "login.html",
        {"error": "Invalid username or password."},
        status_code=401,
    )

```

```

@router.get("/dashboard")
def dashboard(request: Request):
    if not session_is_active(request):
        return RedirectResponse("/login", status_code=303)

```

```

    return templates.TemplateResponse(
        request,
        "dashboard.html",
        {"name": request.session.get("name", DISPLAY_NAME)},
    )

@router.get("/logout")
def logout(request: Request):
    request.session.clear()
    return RedirectResponse("/", status_code=303)

```

1 para explanation of part 1:

The FastAPI application uses Starlette SessionMiddleware to maintain authentication state. Successful login stores the authenticated user's information and a `last_activity` timestamp in the session. The `/dashboard` route checks the session before displaying protected content. Sessions that have been logged out or idle for more than 60 seconds are cleared and redirected to `/login`. Bootstrap is used to style the home, login, and dashboard pages.

PART2 -

For the retrieval experiment, I used a local corpus of six public documents related to my assigned domain, Rental Housing Listings. The corpus contains documents from the California Department of Real Estate, U.S. Department of Housing and Urban Development, Consumer Financial Protection Bureau, and Federal Trade Commission. The total corpus size is 9,826,091 bytes, which exceeds the required 200 KB minimum. Five questions and their expected source documents were defined in `questions.yaml` and committed before retrieval results were generated.

| Technique                  | Configuration                    |
|----------------------------|----------------------------------|
| TokenTextSplitter          | chunk_size=512, chunk_overlap=64 |
| SemanticSplitterNodeParser | buffer_size=1                    |
| SentenceWindowNodeParse    | window_size=3                    |
| r                          |                                  |

|                 |                                        |
|-----------------|----------------------------------------|
| Embedding Model | sentence-transformers/all-MiniLM-L6-v2 |
| Top-k           | 5                                      |

### Figure 10. TokenTextSplitter Retrieval Output for Question q3

The TokenTextSplitter pipeline retrieved the top five chunks for q3 using the 384-dimensional query embedding. The output reports the vector-store score, independently calculated cosine similarity, chunk length, source document, and text preview for each retrieved chunk

```
SCREENSHOT TokenTextSplitter q3
Technique: TokenTextSplitter
Question: According to HUD's rental applicant screening guidance, what share of eviction cases are filed against Black renters, and which eviction records should housing providers and screening companies avoid?
Expected answer: Even though fewer than one in five renters are Black, over half of all eviction cases are filed against Black renters. Housing providers and tenant screening companies should not rely on eviction records that are old, incomplete, or irrelevant, or where a better measure of the applicant's behavior is available.
Expected source: hud_rental_screening_guidance.pdf
Embedding model: sentence-transformers/all-MiniLM-L6-v2
Query embedding dimension: 384
First 8 embedding values: [-0.013543, 0.057094, -0.029385, 0.0459, 0.03562, 0.047089, -0.036552, -0.143655]
Query vector shape: [384]
Document matrix shape: [5, 384]
Top-k: 5
Retrieval latency ms: 10.79
rank | store_score | cosine_similarity | chunk_length | source | preview
1 | 0.828425 | 0.779636 | 1079 | hud_rental_screening_guidance.pdf | see, e.g., also Hepburn, Renee Louis & Matthew Desmond, Racial and Gender Disparities Among Evicted Americans, 7 SOC. SCI. 649, 658 (2020). April 29, 2024 20 groups bearing even higher disparities.
106 Evictions are filed against adults living
2 | 0.792697 | 0.65735 | 984 | hud_rental_screening_guidance.pdf | evictions are notably unreliable: 22% of the eviction records evaluated in a large study contained ambiguous information on how the case was resolved or falsely represented a tenant's eviction history.
104 Eviction disproportionately affects
3 | 0.784372 | 0.662168 | 1211 | hud_rental_screening_guidance.pdf | chance afterwards to have the record disregarded and corrections made (as discussed in Sections V.A.2 and V.B.5 above). Note that under the Fair Credit Reporting Act, tenant screening companies must include existing disposition information
4 | 0.769398 | 0.658011 | 944 | hud_rental_screening_guidance.pdf | an applicant who has since begun receiving rental assistance (e.g. from a government agency) will pay rent on time. In certain circumstances, it would be particularly problematic to hold a past eviction against a tenant, for example an eviction
5 | 0.751671 | 0.675653 | 1242 | hud_rental_screening_guidance.pdf | For example, poor credit could be caused by a gap between a loss of employment income and receipt of disability benefits. Having no credit history or limited credit history (i.e., an absence of credit history, negative or positive) is even
```

### Figure 11. SemanticSplitterNodeParser Retrieval Output for Question q3

The semantic chunking pipeline retrieves larger semantically grouped passages. The output shows the same query and metrics used for the token-based experiment, allowing the techniques to be compared using identical retrieval conditions.

```

Technique: SemanticSplitterNodeParser
Question: According to HUD's rental applicant screening guidance, what share of eviction cases are filed against Black renters, and w
hich eviction records should housing providers and screening companies avoid?
Expected answer: Even though fewer than one in five renters are Black, over half of all eviction cases are filed against Black renter
s. Housing providers and tenant screening companies should not rely on eviction records that are old, incomplete, or irrelevant, or w
here a better measure of the applicant's behavior is available.
Expected source: hud_rental_screening_guidance.pdf
Embedding model: sentence-transformers/all-MiniLM-L6-v2
Query embedding dimension: 384
First 8 embedding values: [-0.013543, 0.057094, -0.029385, 0.0459, 0.03562, 0.047089, -0.036552, -0.143655]
Query vector shape: [384]
Document matrix shape: [5, 384]
Top-k: 5
Retrieval latency ms: 7.17
rank | store_score | cosine_similarity | chunk_length | source | preview
1 | 0.739907 | 0.683133 | 3662 | hud_rental_screening_guidance.pdf | 12, 2023). April 29, 2024 21 experience of domestic violence, da
ting violence, sexual assault, or stalking (e.g., for related noise violations) should not be held against the survivor.110 In additi
on, reasonable accommodations to the screen
2 | 0.723941 | 0.708351 | 7146 | cfpb_tenant_background_market_report.pdf | 1344. 119 MLRI, "Evicted for Life," supra note 112, p. 4-
8. 120 Elora Raymond et al., "Corporate Landlords, Institutional Investors, and Displacement: Eviction Rates in Single-Family Rental
s," Federal Reserve Bank of Atlanta, December 2016
3 | 0.714222 | 0.709219 | 4009 | cfpb_tenant_background_market_report.pdf | 4. CONSUMER FINANCIAL PROTECTION BUREAU 33 homelessness,
causing mental and physical health issues, and preventing equitable wealth-building opportunities.139 As noted in Section 2, the mak
e-up of renters skews towards lower-income famili
4 | 0.659304 | 0.632777 | 3762 | hud_rental_screening_guidance.pdf | 49 15 U.S.C. §§ 1681-1681x; see generally Advisory Opinion, Fair
Credit Reporting; Background Screening, 89 Fed. Reg. 4171 (Jan. 23, 2024). 50 See Langolis v. Abington Hous. Auth., 234 F. Supp. 2d
33, 69 (D. Mass. 2002) ("If the justificati
5 | 0.654629 | 0.584315 | 2840 | hud_rental_screening_guidance.pdf | 649, 667 (2021). April 29, 2024 10 remaining factors, thereby re
creating the same discriminatory outcomes.56 Monitoring should also look at whether any datasets used are more inaccurate or incomple
te for certain groups than others. It also

```

**Figure 12. SentenceWindowNodeParser Retrieval Output for Question q3**

The sentence-window pipeline splits the corpus into individual sentence-level nodes while retaining neighboring context in metadata. The same q3 query and top-5 retrieval setting were used for comparison with the other techniques.

```

SCREENSHOT SentenceWindowNodeParser q3
Technique: SentenceWindowNodeParser
Question: According to HUD's rental applicant screening guidance, what share of eviction cases are filed against Black renters, and w
hich eviction records should housing providers and screening companies avoid?
Expected answer: Even though fewer than one in five renters are Black, over half of all eviction cases are filed against Black renter
s. Housing providers and tenant screening companies should not rely on eviction records that are old, incomplete, or irrelevant, or w
here a better measure of the applicant's behavior is available.
Expected source: hud_rental_screening_guidance.pdf
Embedding model: sentence-transformers/all-MiniLM-L6-v2
Query embedding dimension: 384
First 8 embedding values: [-0.013543, 0.057094, -0.029385, 0.0459, 0.03562, 0.047089, -0.036552, -0.143655]
Query vector shape: [384]
Document matrix shape: [5, 384]
Top-k: 5
Retrieval latency ms: 45.95
rank | store_score | cosine_similarity | chunk_length | source | preview
1 | 0.849944 | 0.698302 | 319 | hud_rental_screening_guidance.pdf | Even though fewer than one in five renters are Black, over half o
f all eviction cases are filed against Black renters.105 Black and Hispanic renters have eviction cases filed against them at higher
rates than White renters, with women of th
2 | 0.813259 | 0.765805 | 813 | hud_rental_screening_guidance.pdf | Eviction History Eviction records are one of the most commonly-ma
rketed tenant screening components, and they are standard features of many screening reports.101 Tenant screening companies have buil
t private databases from court records of
3 | 0.801946 | 0.727957 | 205 | hud_rental_screening_guidance.pdf | Therefore, tenant screening companies and housing providers shou
ld not rely on eviction records that are old, incomplete, irrelevant, or where a better measure of an applicant's behavior is availabl
e.
4 | 0.790346 | 0.685743 | 447 | hud_rental_screening_guidance.pdf | April 29, 2024 20 groups bearing even higher disparities.106 Evic
tions are filed against adults living with children at more than twice the rate than for adults living without children.107 Some grou
ps of renters with disabilities disproport
5 | 0.758575 | 0.686312 | 377 | hud_rental_screening_guidance.pdf | Note that under the Fair Credit Reporting Act, tenant screening c
ompanies must include existing disposition information for eviction records.109 Housing providers and tenant screening companies shou
ld not deny or recommend the denial of hou

```

| Technique | Chunks | Avg chunk length | Top-1 cosine | Mean@5 cosine | Recall@5 | Mean retrieval latency (ms) |
|-----------|--------|------------------|--------------|---------------|----------|-----------------------------|
|-----------|--------|------------------|--------------|---------------|----------|-----------------------------|

|                            |      |         |          |          |     |       |
|----------------------------|------|---------|----------|----------|-----|-------|
| TokenTextSplitter          | 839  | 1042.78 | 0.673236 | 0.645124 | 1.0 | 12.14 |
| SemanticSplitterNodeParser | 286  | 2676.38 | 0.575736 | 0.565879 | 1.0 | 7.48  |
| SentenceWindowNodeParser   | 5538 | 138.22  | 0.747511 | 0.646602 | 1.0 | 47.62 |

All three techniques achieved a Recall@5 of 1.0, meaning that the expected source document appeared somewhere within the top five retrieved results for all five questions. However, the techniques differed substantially in chunk count, similarity scores, and retrieval latency. Sentence-window chunking produced the highest average top-1 cosine similarity at 0.747511, but it also generated 5,538 chunks and had the highest mean retrieval latency at 47.62 ms. Semantic chunking generated only 286 chunks and had the lowest mean retrieval latency at 7.48 ms, but it produced the lowest top-1 and mean@5 cosine similarities. Token chunking provided an intermediate result with 839 chunks, 12.14 ms mean latency, and similarity scores close to sentence-window at the top-5 level.

For this experiment, Recall@5 is 1 when at least one of the five retrieved chunks comes from the expected source file recorded in questions.yaml, and 0 otherwise. The final Recall@5 for each technique is the average across the five questions.

Confidently Scored Retrieval That Did Not Contain the Answer:

A useful false-positive example occurred for q3 with SemanticSplitterNodeParser. The second-ranked chunk received a store score of 0.723941 and an independently calculated cosine similarity of 0.708351, but it came from cfpb\_tenant\_background\_market\_report.pdf instead of the expected hud\_rental\_screening\_guidance.pdf.

The CFPB chunk discusses eviction filings, tenant-screening reports, sealed or ambiguous eviction records, and inaccurate eviction information. These concepts are semantically very close to q3, which asks about eviction cases and the types of eviction records that housing providers should avoid. As a result, the embedding model assigned the chunk a high similarity score even though it did not contain the specific HUD facts required to answer the question.

This example shows that a high cosine similarity indicates semantic relatedness, but does not guarantee that the retrieved passage contains the exact answer.

Conclusion:

For this corpus, I would choose TokenTextSplitter as the most balanced technique. All three methods achieved Recall@5 of 1.0, but token chunking maintained a mean@5 cosine similarity

nearly identical to sentence-window chunking while using far fewer chunks and substantially lower retrieval latency. Sentence-window achieved stronger top-1 similarity, while semantic chunking was the fastest, so the preferred method can change depending on whether retrieval precision, storage/index size, or latency is prioritized.

AI use:

**1. What did you use an AI assistant for and what did you do yourself?**

I used an AI assistant mainly to help me understand the homework requirements, explain concepts while I was working, debug code when I ran into issues, and figure out what screenshots and evidence were needed for the report. I ran the application, tested the authentication flow, ran the retrieval experiments, collected the screenshots, and verified the outputs myself.

**2. Give one AI-produced output that was wrong/unsuitable, or one thing you independently verified.**

One thing I independently verified was the average chunk-length calculation. The first version calculated the average using only the retrieved top-5 chunks instead of all chunks produced by each chunking technique.

**3. How did you detect the problem or verify the result?**

I checked the calculation against the homework instructions and noticed that the assignment asked for the average length of all chunks produced by each chunker. I then reviewed how the metric was being calculated and confirmed that it needed to use the full set of chunks.

**4. What did you change and why does it work now?**

I changed the calculation so that average chunk length is computed across every node produced by each chunking technique. The corrected values were saved in `chunk_stats.json`, and the summary script now uses those full-index statistics when generating the final metrics table.